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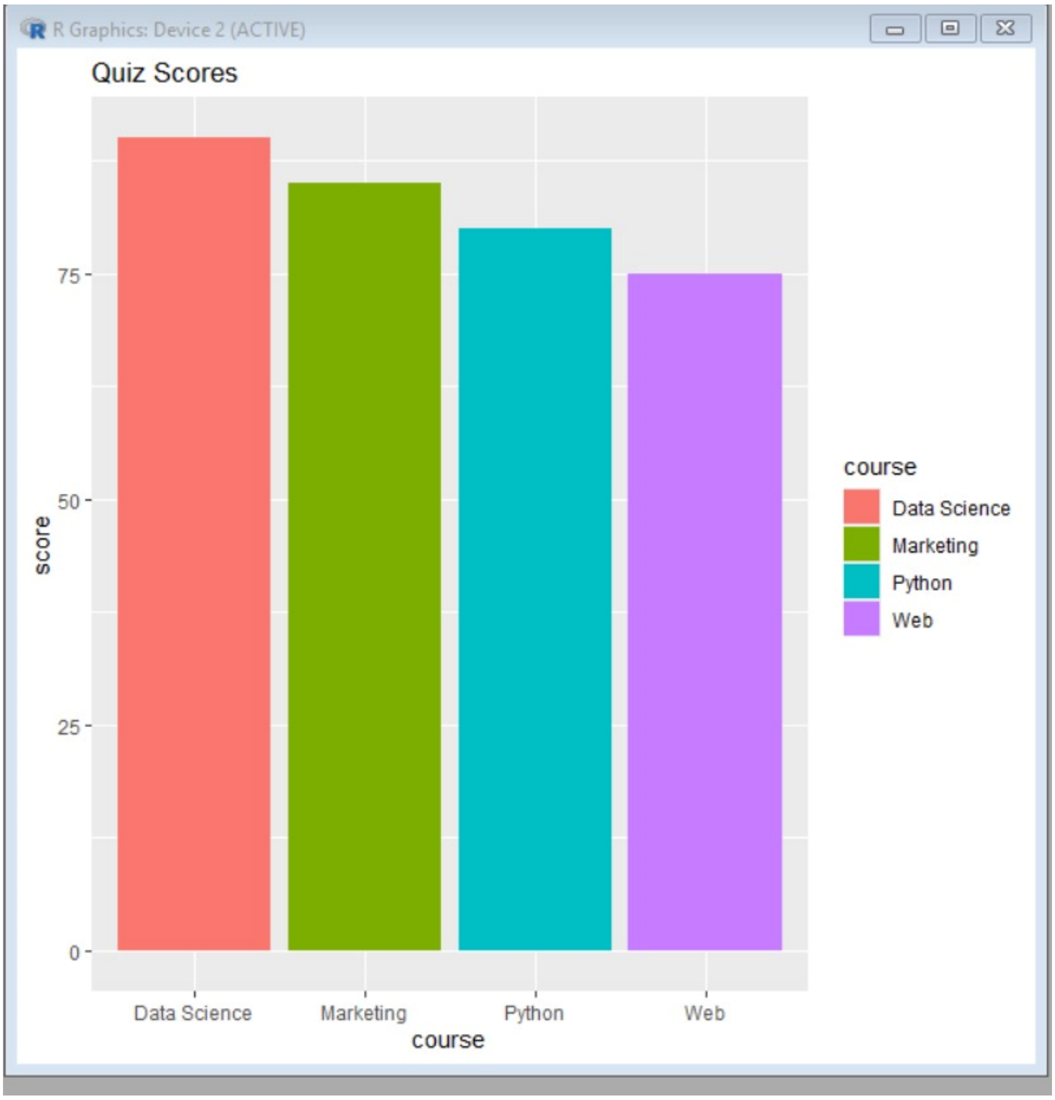
COURSE CODE :DSA0604

SCENARIO 1:

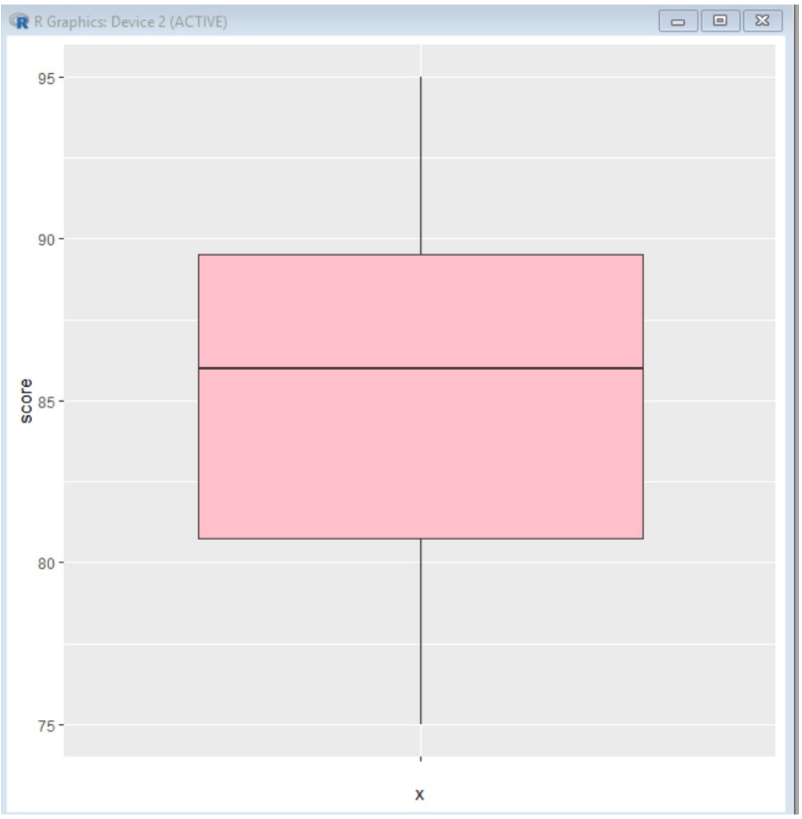
Untitled - R Editor

```r
# Load library
library(ggplot2)

# Create dataset
student <- data.frame(
  course = c("Python","Data Science","Web","Marketing"),
  score = c(80,90,75,85)
)

# View data
print(student)

# Bar Chart
ggplot(student, aes(x=course, y=score, fill=course)) +
  geom_bar(stat="identity") +
  labs(title="Quiz Scores")
```


R Graphics: Device 2 (ACTIVE)
Quiz Scores
score
0
25
50
75
Data Science
Marketing
Python
Web
course
course
Data Science
Marketing
Python
Web

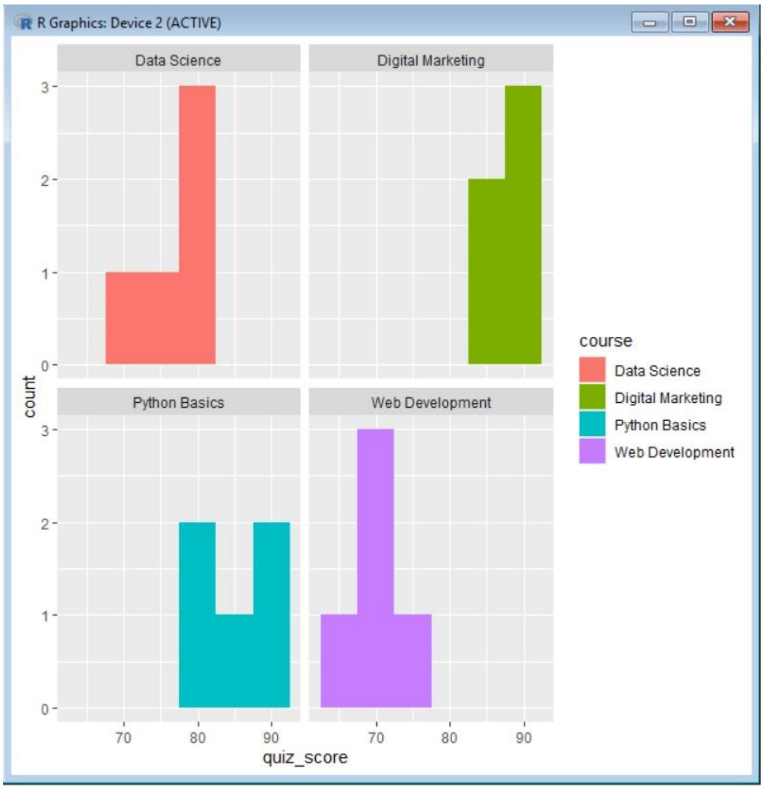
SCENARIO 2:

```r
# Load library
library(ggplot2)

# Python Basics quiz scores
score <- c(78, 85, 92, 88, 75, 80, 95, 83, 90, 87)

# 1. Histogram (Binwidth = 5)
ggplot(data.frame(score), aes(score)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black")

# Histogram (Binwidth = 10)
ggplot(data.frame(score), aes(score)) +
  geom_histogram(binwidth = 10, fill = "orange", color = "black")

# 2. Density Plot
ggplot(data.frame(score), aes(score)) +
  geom_density(fill = "lightgreen")

# 3. Mean, Median, Standard Deviation
mean(score)
median(score)
sd(score)

# 4 & 5. Box Plot (Shape & Outliers)
ggplot(data.frame(score), aes(x = "", y = score)) +
  geom_boxplot(fill = "pink")
```

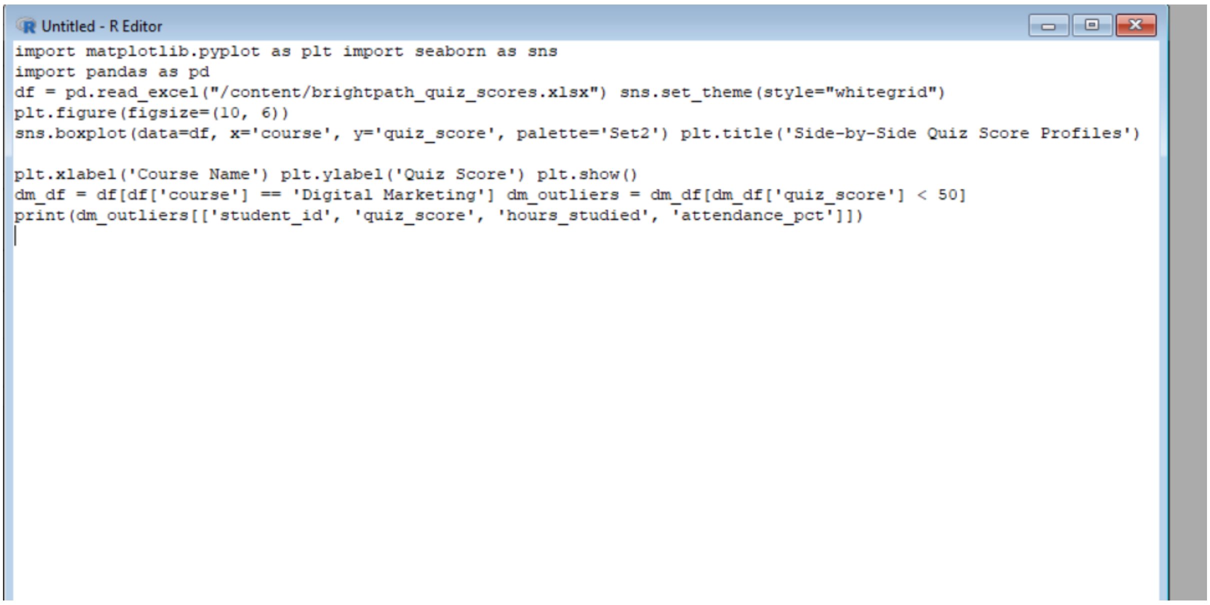


SCENARIO 3:

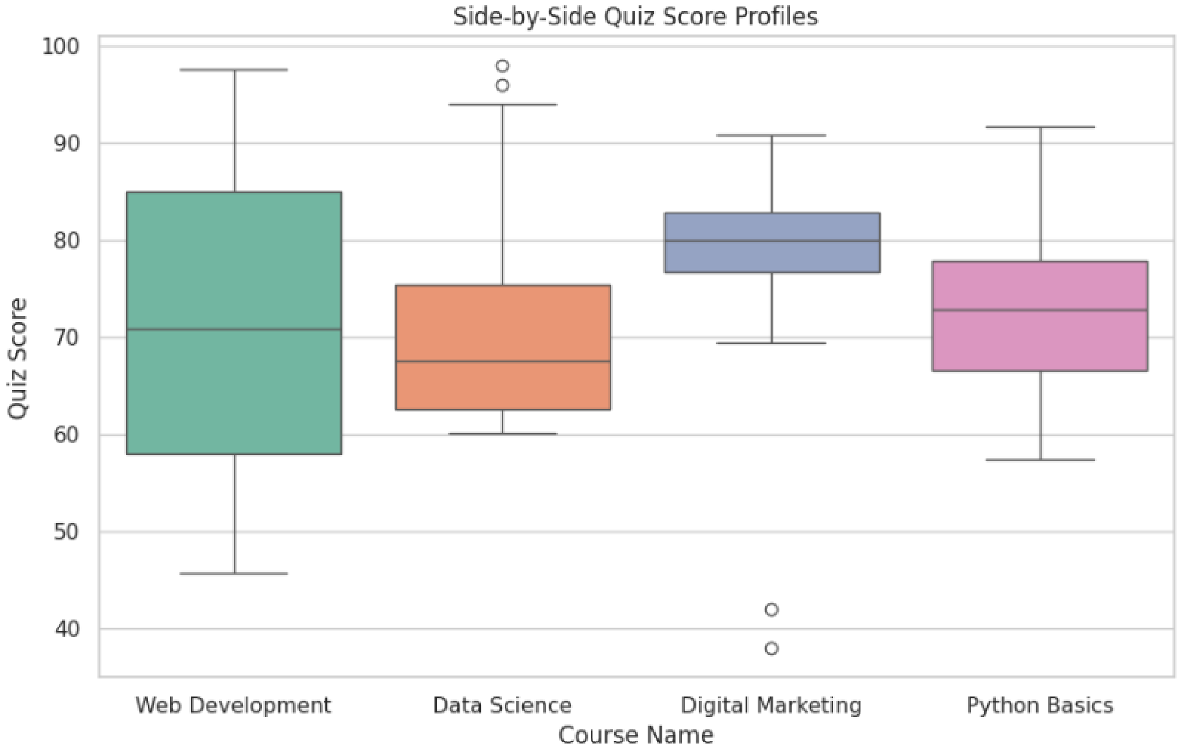
```
# Load library
library(ggplot2)

# Dataset
student <- data.frame(
  course = rep(c("Python Basics","Data Science","Web Development","Digital Ma
  quiz_score = c(78,85,92,88,80,
                 70,75,82,78,80,
                 65,68,72,75,70,
                 85,88,90,92,87)
)

# 1. Overlaid Density Plot
ggplot(student, aes(x=quiz_score, color=course)) +
  geom_density()

# 2. Histogram for Each Course
ggplot(student, aes(x=quiz_score, fill=course)) +
  geom_histogram(binwidth=5) +
  facet_wrap(~course)

# 3. Summary Statistics
aggregate(quiz_score ~ course, student, mean)
aggregate(quiz_score ~ course, student, sd)
```



SCENARIO 4:




```
import matplotlib.pyplot as plt import seaborn as sns
import pandas as pd
df = pd.read_excel("/content/brightpath_quiz_scores.xlsx") sns.set_theme(style="whitegrid")
plt.figure(figsize=(10, 6))
sns.boxplot(data=df, x='course', y='quiz_score', palette='Set2') plt.title('Side-by-Side Quiz Score Profiles')

plt.xlabel('Course Name') plt.ylabel('Quiz Score') plt.show()
dm_df = df[df['course'] == 'Digital Marketing'] dm_outliers = dm_df[dm_df['quiz_score'] < 50]
print(dm_outliers[['student_id', 'quiz_score', 'hours_studied', 'attendance_pct']])
```




```
     student_id  quiz_score  hours_studied  attendance_pct
34       BP4000        38.0            6.8            84.1
111      BP4001        42.0           10.2            79.8
```

SCENARIO 5:

```r
# Load library
library(ggplot2)

# Dataset
student <- data.frame(
  course = rep(c("Python Basics","Data Science",
                 "Web Development","Digital Marketing"), each = 5),
  quiz_score = c(78,85,92,88,80,
                 70,75,82,78,80,
                 65,68,72,75,70,
                 85,88,90,92,87)
)

# 1. Side-by-side Boxplots
ggplot(student, aes(x = course, y = quiz_score, fill = course)) +
  geom_boxplot() +
  labs(title = "Quiz Score by Course")

# 2. Median
aggregate(quiz_score ~ course, student, median)

# 3. IQR
aggregate(quiz_score ~ course, student, IQR)

# 4. Summary (Outliers if any)
by(student$quiz_score, student$course, summary)
```